

The arbitrary-measure isomorphic Busemann–Petty constant has optimal order $\sqrt{n}$

Candidate full solution packet for arXiv:2305.17796

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Status. candidate full, likely valid; requires human review.

1 Source question

Koldobsky, Roysdon, and Zvavitch ask on page 5 of *Comparison Problems for Radon Transforms* [1] whether the $\sqrt{n}$ constant in the arbitrary-measure isomorphic Busemann–Petty inequality is optimal. Their statement is reproduced below.

ratio distance. In the case when f is even, K is origin-symmetric and convex, we may apply John’s theorem [24] to conclude that $d_{\text{ovr}}(K, \mathcal{I}_n) \leq \sqrt{n}$.

An isomorphic version of the measure theoretic Busemann–Petty problem from [51] was proved in [39]: Given a non-negative, continuous function $f: \mathbb{R}^n \rightarrow \mathbb{R}_+$, and a pair of origin-symmetric convex bodies K, L in $\mathbb{R}^n$ satisfying $\int_{K \cap \theta^\perp} f \leq \int_{L \cap \theta^\perp} f$ for every $\theta \in S^{n-1}$, one has

$$(7) \quad \int_K f \leq \sqrt{n} \int_L f.$$

It is still an open problem to determine whether the constant $\sqrt{n}$ is optimal. In [15] the following extension of the inequality (7) was established: Let K and L be star bodies in $\mathbb{R}^n$ and let $f, g: \mathbb{R}^n \rightarrow \mathbb{R}_+$ be non-negative, continuous functions on K and L , respectively, so that $\|g\|_\infty = g(0) = 1$. Then

$$(8) \quad \int_K f \leq \frac{n}{n-1} d_{\text{ovr}}(K, \mathcal{I}_n) \max_{\theta \in S^{n-1}} \left(\frac{\int_{K \cap \theta^\perp} f}{\int_{L \cap \theta^\perp} g} \right) |K|^{\frac{1}{n}} \left(\int_L g \right)^{\frac{n-1}{n}}.$$

For the current state of the Busemann–Petty and slicing problems for functions see the survey [16].

Let C_n be the least number with the following property. For every non-negative continuous even function $f: \mathbb{R}^n \rightarrow \mathbb{R}_+$ and origin-symmetric convex bodies $K, L \subset \mathbb{R}^n$, the inequalities

$$\int_{K \cap \xi^\perp} f d\mathcal{H}^{n-1} \leq \int_{L \cap \xi^\perp} f d\mathcal{H}^{n-1} \quad (\xi \in S^{n-1}) \tag{1}$$

imply

$$\int_K f dx \leq C_n \int_L f dx. \tag{2}$$

Koldobsky and Zvavitch proved $C_n \leq \sqrt{n}$ [2].

2 Idea of the proof

Klartag and Livshyts constructed a probability density whose mass inside a convex body A is bounded below, although its mass on every affine hyperplane is only $O(n^{-1/2})$ [3]. Their explicit

construction has two geometric features crucial here: A has bounded volume radius and contains an absolute multiple of $\sqrt{n}B_2^n$, while the Gaussian components are centered at distance cn from the origin.

Fix a sufficiently large absolute R and put $D = R\sqrt{n}B_2^n$. A fixed positive amount of the slicing measure lies in $A \setminus D$. The bounded volume radius and the inball imply, by taking two cones over any central section of A , that $|A \cap H| \leq C\sqrt{n}|A|$. For fixed large R , $D \setminus A$ therefore occupies a fixed fraction of every central section of D .

Keep the slicing density only on $A \setminus D$ and place a second, tiny continuous density on $D \setminus A$. The supports are disjoint. Scaling the second density makes its section in every H dominate the first one. Yet the second total mass is at most

$$\frac{C}{\sqrt{n}} \frac{|D|}{|D \cap H|} = O(n^{-1/2}),$$

whereas the first mass is bounded below. Their sum is one common continuous even density.

3 Formal result

Theorem 1. *There is a universal constant $c > 0$ such that, for every $n \geq 2$,*

$$c\sqrt{n} \leq C_n \leq \sqrt{n}.$$

Thus the $\sqrt{n}$ estimate in (2) is optimal in its dimension dependence.

The upper bound is [2]; we prove the lower bound. Write $\omega_m = |B_2^m|$.

4 A positioned sharp slicing example

Lemma 2 (positioned Klartag–Livshyts example). *There are absolute constants $a, b, c_0, M > 0$ such that, for all sufficiently large n , there are an origin-symmetric convex body $A \subset \mathbb{R}^n$ and an even probability measure ν with continuous density p satisfying*

$$a\sqrt{n}B_2^n \subset A, \quad |A| \leq b^n, \quad (3)$$

$$\sup_{\xi \in S^{n-1}, t \in \mathbb{R}} \int_{\xi^\perp + t\xi} p d\mathcal{H}^{n-1} \leq \frac{M}{\sqrt{n}}. \quad (4)$$

Moreover, for every fixed $R > 0$, after increasing the threshold on n ,

$$\nu(\text{int}(A) \setminus R\sqrt{n}B_2^n) \geq c_0. \quad (5)$$

Proof. Recall the construction in Section 4 of [3]. Their body before the final absolute dilation is

$$K_0 = \text{conv}\{\pm R_k \theta_{kj}, \pm ne_i : 1 \leq k \leq m\}, \quad A = C_6 K_0.$$

Their displayed estimates (1) and (2) prove $|A| \leq b^n$ and (4) for a suitable realization. Since $nB_1^n = \text{conv}\{\pm ne_i\} \supset \sqrt{n}B_2^n$, the inball in (3) follows.

The measure ν is the average of standard Gaussian translates centered at

$$z = \pm \sum_{k=1}^m \tilde{R}_k \theta_{kj_k}, \quad \tilde{R}_k = \frac{n}{(\log^{(k)} n)^{3/2}}. \quad (6)$$

The universal constant in the construction may harmlessly be enlarged so the defining separation condition gives $\tilde{R}_k/\tilde{R}_{k-1} \geq 3$ for $2 \leq k \leq m$. Also, equation (4.1) of [3] gives $\log^{(m)} n \leq C'_0$. Consequently

$$|z| \geq \tilde{R}_m - \sum_{k < m} \tilde{R}_k \geq \frac{1}{2} \tilde{R}_m \geq \frac{n}{2(C'_0)^{3/2}} =: c_1 n. \quad (7)$$

This is deterministic for every center.

The proof of equation (3) in [3] observes that every center belongs to $A/2$ and $2\sqrt{n}B_2^n \subset A/2$. If G is standard Gaussian, then on $\{|G| < 2\sqrt{n}\}$ one has $z + G \in \text{int}(A)$. For fixed R and large n , (7) also gives $|z + G| > R\sqrt{n}$. Finally, $\mathbb{P}(|G| < 2\sqrt{n}) \geq 3/4$ by Markov applied to $|G|^2$. Thus one may take $c_0 = 1/2$ after ignoring finitely many dimensions. $\square$

5 A ball with a uniformly thick complement

Lemma 3. *If A satisfies (3), there is an absolute $R > 0$, depending only on a, b , such that for large n , with $D = R\sqrt{n}B_2^n$,*

$$|(D \setminus A) \cap \xi^\perp| \geq \frac{1}{2} |D \cap \xi^\perp| \quad (\xi \in S^{n-1}). \quad (8)$$

Proof. Fix $H = \xi^\perp$. The points $\pm a\sqrt{n}\xi$ belong to A . The two cones in A with base $A \cap H$ and these apices have disjoint interiors, so

$$|A| \geq 2 \frac{a\sqrt{n}}{n} |A \cap H|, \quad |A \cap H| \leq \frac{\sqrt{n}}{2a} b^n. \quad (9)$$

There is an absolute $c > 0$ with $\omega_{n-1} \geq (c/\sqrt{n})^{n-1}$. Hence $|D \cap H| = (R\sqrt{n})^{n-1} \omega_{n-1} \geq (cR)^{n-1}$. Choose fixed R so large that this is at least twice the right side of (9) for all large n . Since $|(A \cap D) \cap H| \leq |A \cap H|$, subtraction proves (8). $\square$

6 The common-density separation construction

Lemma 4. *For A, ν, p, D as above there is a non-negative continuous even density f such that*

$$\int_{A \cap \xi^\perp} f d\mathcal{H}^{n-1} \leq \int_{D \cap \xi^\perp} f d\mathcal{H}^{n-1} \quad (\xi \in S^{n-1}), \quad (10)$$

but, for absolute $c_2, C_2 > 0$,

$$\int_A f dx \geq c_2, \quad \int_D f dx \leq \frac{C_2}{\sqrt{n}}. \quad (11)$$

Proof. Put $U = \text{int}(A) \setminus D$. By (5) and inner regularity, there is an even continuous $0 \leq \chi \leq 1$, vanishing outside U , such that

$$g := p\chi, \quad G := \int_{\mathbb{R}^n} g \geq c_0/2. \quad (12)$$

As $g \leq p$, (4) gives

$$s := \sup_{\xi \in S^{n-1}} \int_{\xi^\perp} g d\mathcal{H}^{n-1} \leq \frac{M}{\sqrt{n}}. \quad (13)$$

Let $V = \text{int}(D) \setminus A$ and $h_j(x) = \min\{1, j \text{dist}(x, V^c)\}$. These functions are continuous and even, vanish on A and outside D , and increase pointwise to 1_V . Their central Radon transforms $F_j(\xi) = \int_{\xi^\perp} h_j d\mathcal{H}^{n-1}$ are continuous and increase to

$$F(\xi) = |V \cap \xi^\perp| = |D \cap \xi^\perp| - |(A \cap D) \cap \xi^\perp|.$$

The limit is continuous because central-section volume varies continuously with direction for each convex body D and $A \cap D$. Dini's theorem makes the convergence uniform. By (8), for some j the function $h = h_j$ satisfies

$$q := \inf_{\xi \in S^{n-1}} \int_{\xi^\perp} h d\mathcal{H}^{n-1} \geq \frac{1}{4} (R\sqrt{n})^{n-1} \omega_{n-1}. \quad (14)$$

Set $\lambda = s/q$ and $f = g + \lambda h$. Since g vanishes on D and h on A , for every ξ ,

$$\int_{A \cap \xi^\perp} f = \int_{\xi^\perp} g \leq s = \lambda q \leq \lambda \int_{\xi^\perp} h = \int_{D \cap \xi^\perp} f.$$

Also $\int_A f = G \geq c_0/2$. Using $0 \leq h \leq 1$, (13), and (14),

$$\begin{aligned} \int_D f &= \lambda \int_D h \leq \frac{M}{\sqrt{n}} \frac{|D|}{q} \\ &\leq \frac{4M}{\sqrt{n}} \frac{(R\sqrt{n})^n \omega_n}{(R\sqrt{n})^{n-1} \omega_{n-1}} = 4MR \frac{\omega_n}{\omega_{n-1}} \leq \frac{C_2}{\sqrt{n}}, \end{aligned}$$

where the last inequality is the standard gamma-ratio estimate $\omega_n/\omega_{n-1} \leq C/\sqrt{n}$. $\square$

Proof of Theorem 1. For all large n , Lemma 4 gives two origin-symmetric convex bodies A, D and a common density f obeying the section condition, while

$$\frac{\int_A f}{\int_D f} \geq c\sqrt{n}.$$

Thus $C_n \geq c\sqrt{n}$ in all large dimensions. Also $C_n \geq 1$ in every dimension, by taking the bodies equal and a density of positive mass. Decreasing c absorbs the finitely many remaining dimensions. Combine this with $C_n \leq \sqrt{n}$ from [2]. $\square$

7 Verification notes and limitations

- The proof uses the explicit centers and body in Section 4 of [3], not only the abstract slicing theorem.
- No discontinuous density is used. The cutoff on $D \setminus A$ is uniform in direction by Dini's theorem.
- The density is non-negative, continuous, and even. It may vanish, as the source formulation permits.
- The result determines the optimal order, not the exact numerical value of C_n .
- The accompanying script checks the unit-ball gamma ratio and final $n^{-1/2}$ scaling numerically, but is not a proof dependency.

8 Bounded novelty search

On 21 July 2026 we searched the local 2000–2026 arXiv source corpus and the web for the exact question and close variants, including `isomorphic Busemann-Petty same measure sqrt n optimal` and `arbitrary measures sharp lower bound`. The search covered the source survey [1], original upper bound [2], two-measure comparison paper [4], and sharp slicing construction [3]. The two-measure version has sharp $\sqrt{n}$ order [4], but that paper explicitly leaves the same-measure constant open. We found no later statement of the lower bound proved here. Novelty confidence is moderate rather than definitive.

9 Human-review recommendation

Check: (i) that the constants in Section 4 of [3] may be enlarged so consecutive $\tilde{R}_k$ grow by at least 3; (ii) the two-cone estimate (9); and (iii) continuity of the central-section function of $A \cap D$, used in Dini’s theorem. Once these pass, the rest is direct support separation and scaling.

References

- [1] A. Koldobsky, M. Roysdon, and A. Zvavitch, *Comparison Problems for Radon Transforms*, Adv. Math. 434 (2023), 109336; arXiv:2305.17796.
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- [3] B. Klartag and G. V. Livshyts, *The lower bound for Koldobsky’s slicing inequality via random rounding*, Geometric Aspects of Functional Analysis, LNM 2256 (2020), 43–57; arXiv:1810.06189.
- [4] A. Koldobsky, G. Paouris, and A. Zvavitch, *Measure comparison and distance inequalities for convex bodies*, Indiana Univ. Math. J. 71 (2022), 391–407; arXiv:1912.00880.